

WANG HOUTIAN

Undergraduate Physics Student · Nanyang Technological University

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EDUCATION

Nanyang Technological University, Singapore

Expected 2028

Bachelor of Science in Physics

CGPA: 4.38/5.00

Transferred from Materials Science and Engineering to Physics in 2026.

RESEARCH EXPERIENCE

Memory Effects in Stochastic Thermodynamics

2026–Present

Undergraduate Research Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Investigating memory effects in stochastic thermodynamics through a variational formulation proposed by Prof. François Gay-Balmaz.
- Applying the formulation to the full non-Markovian system, with emphasis on fluctuation–dissipation consistency, stationary behavior, and entropy production.
- Developing analytical derivations for stochastic systems with memory and checking their thermodynamic consistency.

RESEARCH INTERESTS

Undergraduate focus: Non-Markovian stochastic dynamics; stochastic thermodynamics; nonequilibrium statistical mechanics.

Future directions: Mathematical physics, symmetry, and representations; quantum field theory, holography, and AdS/CFT; quantum information theory.

SELECTED ACADEMIC WRITING

Analytical Mechanics

Available on website

Structured notes on Lagrangian and Hamiltonian mechanics.

Classical Field Theory

In preparation

Quantum Mechanics

In preparation